

VertexLearn AI — Simple Project Action Guide

Intern: **Ali Charolia** • Front-End Internship • Internmo Submission Guide

FRONT-END TRACK



The 5 Screens You Need to Build

1. Course Catalog & Dashboard

Browse courses and view progress.

- Search bar & category filter buttons (Web Dev, Python, AI).
- Course cards with thumbnail, title, and "Enroll" button.
- "My Learning" showing progress bar (%) & daily streak.

2. Video Player & Learning Page

Lecture watching interface.

- Video player in center (embed sample YouTube or MP4).
- Lesson playlist on right (click to switch video).
- Personal Notes box below video (auto-saves to LocalStorage).

3. Embedded AI Tutor (Star Feature)

Interactive chat drawer beside video.

- **Chat Box:** AI answers with lesson timestamp (e.g. "See 03:15").
- **Difficulty Toggle:** Beginner | Intermediate | Advanced.
- **Quick Buttons:** "Summarize Lesson" & "Show Flashcards".

4. Quiz & Certificate Downloader

Assessment & proof of completion.

- 5 Multiple Choice Questions with radio buttons & timer.
- Instant score feedback (e.g. "You scored 4/5! Passed").
- **"Download Certificate"** button (generates PDF diploma for **Ali Charolia**).

5. Instructor & Admin Quick View

Simple management screens for the evaluator.

- **Instructor View:** A simple form to "Add New Course" + basic chart.
- **Admin View:** Table of pending courses with "Approve" and "Reject" buttons.
- **Pro Tip:** Add [View As: Student | Instructor | Admin] in your navbar!



Recommended Simple Tech Stack

Frontend: React + Vite + Tailwind CSS (fast dev, responsive, modern).

Storage: LocalStorage (saves courses, notes, and quiz results locally).

AI Chat: Smart pre-written responses or free Google Gemini API.



Step-by-Step Action Plan (What to do first, second, third)

1

Step 1: Set up the project & sample data

Run the Vite command to create the React app. Create a `mockData.ts` file containing 3–4 sample courses (e.g., "React Essentials", "Python Fundamentals") with lesson titles, video URLs, and quiz questions.

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Step 2: Build Course Catalog & Dashboard

Create the navbar with your logo, search bar, and role switcher. Make course cards with an "Enroll" button. When enrolled, the course appears in "My Learning" with a progress bar.

3

Step 3: Build the Video Player & Notes screen

When clicking a course, open the player page. Put a video on the left and the lesson list on the right. Add a textarea where students type notes that auto-save to `localStorage`.

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Step 4: Build the AI Tutor Chat panel

Add an "AI Tutor" button that opens a chat sidebar. Give it smart mock responses (or connect the free Gemini API) that answer course questions with timestamp citations, plus a button to show 3 flip flashcards.

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Step 5: Add the Quiz modal & Certificate download

Add a "Take Quiz" button with 5 questions. After submitting, show the score. When the course hits 100%, show a "Download Certificate" button that generates a PDF diploma using `jspdf`.

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Step 6: Add Instructor & Admin views & Deploy

Add a simple course creation form and an approval table. Push your code to GitHub and connect it to Vercel (free 1-click deployment).



Final Submission Checklist (For Internmo Portal)

Prepare these 4 items when submitting under "Upload Project File & Submit":

- ✓ **GitHub Repository:** Clean code with a clear `README.md` including screenshots and setup instructions.
- ✓ **Live Website Link:** Deployed on Vercel or Netlify (evaluators test your app directly in their browser).
- ✓ **2-Minute Screen Recording:** Record a quick video showing: catalog → video player → AI tutor chat → taking quiz → downloading certificate.
- ✓ **Submission ZIP File:** A zip file of your project code + a text file with your GitHub and Live Demo links.